Design Patterns

Name: DURGASREE AVVARU

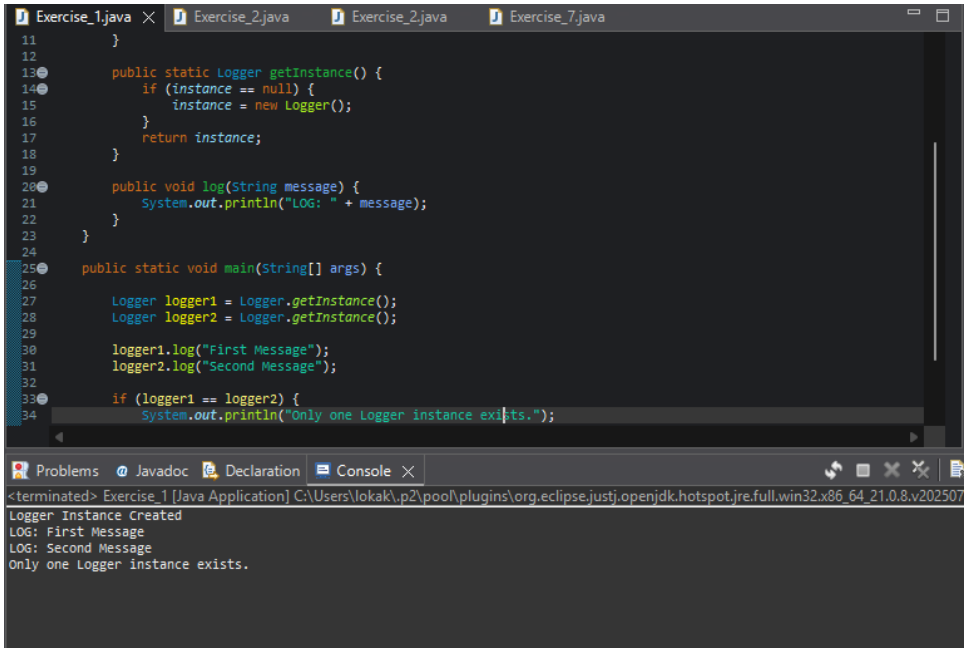
Registered Mail: 2300031591cseh@gmail.com

Exercise 1: Singleton Pattern

```
package Design_patterns;

public class Exercise_1 {

   static class Logger {

      private static Logger instance;

      private Logger() {
         System.out.println("Logger Instance Created");
      }

      public static Logger getInstance() {
         if (instance == null) {
            instance = new Logger();
         }
         return instance;
      }

      public void log(String message) {
         System.out.println("LOG: " + message);
      }
   }

   public static void main(String[] args) {

      Logger logger1 = Logger.getInstance();
      Logger logger2 = Logger.getInstance();

      logger1.log("First Message");
      logger2.log("Second Message");
```

```
        if (logger1 == logger2) {
            System.out.println("Only one Logger instance exists.");
        } else {
            System.out.println("Multiple Logger instances exist.");
        }
    }
}
```

Output Screenshot



Exercise 2: Factory Method Pattern

```
package Design_patterns;

public class Exercise_2 {

    interface Document { void open(); }

    static class WordDocument implements Document {
        public void open() { System.out.println("Word Document Opened"); }
    }
```

```
    static class PdfDocument implements Document {
        public void open() { System.out.println("PDF Document Opened"); }
    }

    static class ExcelDocument implements Document {
        public void open() { System.out.println("Excel Document Opened"); }
    }

    static abstract class DocumentFactory {
        public abstract Document createDocument();
    }

    static class WordFactory extends DocumentFactory {
        public Document createDocument() { return new WordDocument(); }
    }

    static class PdfFactory extends DocumentFactory {
        public Document createDocument() { return new PdfDocument(); }
    }

    static class ExcelFactory extends DocumentFactory {
        public Document createDocument() { return new ExcelDocument(); }
    }

    public static void main(String[] args) {
        DocumentFactory wordFactory = new WordFactory();
        Document wordDoc = wordFactory.createDocument();
        wordDoc.open();

        DocumentFactory pdfFactory = new PdfFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();

        DocumentFactory excelFactory = new ExcelFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
    }
}
```

Output Screenshot: